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SUPERSEDING
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FEDERAL SPECIFICATION

PAPER, ABRASIVE, FLINT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — Flint paper covered by this specification is a paper coated on one surface with a flint abrasive. This flint paper is satisfactory for all ordinary sanding of wood and paint.

1.2 Classification.

1.2.1 *Classes.*—Flint paper covered by this specification shall be of the following classes, as specified:

Class 1.—Sheets, flint finishing paper
Class 2.—Sheets, flint paper
Class 3.—Rolls, flint paper

1.2.2 *Grades.* — Flint paper shall be furnished in the following classes and grades, as specified:

Grade designation

Class 1: 5/0, 4/0, 3/0, and 2/0
Class 2: Extra Fine, Fine, Medium, Coarse, Extra Coarse
Class 3: Extra Fine, Fine, Medium, Coarse, Extra Coarse

NOTE.—The above designations in the order given indicate the finest to the coarser grit size.

1.2.3 *Dimensions of sheets and rolls.*—See 3.3.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following Federal Specifications, of the issues in effect on date of invitation for bids, form a part of this specification:

UU-P-31—Paper; General Specifications and Methods of Testing.
LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).
LLL-B-636—Boxes; Fiber, Solid (for Domestic Shipment).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Abrasive.* — The abrasive shall be crushed or ground from clean hard quartz,

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quartzite, or flint and shall be free from extraneous materials.

3.1.2 *Backing.* — The backing shall be a good quality paper suitable for the purpose intended.

3.1.3 *Adhesive.*—The adhesive shall be a good quality, animal hide glue base, suitable for the purpose intended.

3.2 Coating.

3.2.1 *Class 1.* — The flint finishing paper sheets shall be "open coating". The abrasive flint grain shall be applied to one side of the paper backing only, and cover approximately 50 percent of the backing, thus leaving open spaces between the grains of abrasive.

3.2.2 *Classes 2 and 3.* — The flint paper sheets and rolls shall be "closed coating". The abrasive flint grain shall be applied to one side of the paper backing only, and completely cover the surface of the backing.

3.3 Dimensions of sheets and rolls. — Unless otherwise specified, class 1, flint finishing paper shall be furnished in sheets not less than 9 x 11 inches. Unless otherwise specified, class 2, flint paper shall be furnished in sheets not less than 9 x 10 inches. Class 3, flint paper shall be furnished in rolls not less than 50 yards long, and in the widths, as specified.

TABLE I.—*Weight of paper backing*

Class No.	Grade	Nominal paper weight[1]	Tolerance[2]
		Pounds	*Pounds*
1	All grades, 5/0 thru 2/0	40	37.5– 42.5
2	Extra fine, fine	50 to 60	46.5– 63.0
2	Medium, coarse, extra coarse	55 to 60	51.0– 63.0
3	All grades, extra fine thru extra coarse	130	122.0–135.0

[1] Nominal weights are the approximate weight of 480 sheets, 24 inches by 36 inches.

[2] The backing weight tolerance represents the allowed variation in the weight of 480 sheets, 24 inches by 36 inches, when conditioned at 70° ± 2°F. and 50 ± 2 percent relative humidity for not less than 24 hours.

3.4 Backing.—The backing for class 1, flint finishing paper, and class 3, roll flint paper, shall be a good-quality wood fiber or rope paper. The backing for class 2, flint paper, shall be a good-quality rope, wood fiber, or kraft paper.

3.4.1 *Weight of backing.*—The weight of the paper backing shall be as specified in table I.

3.5 Tensile strength.—The average tensile strength of flint paper per inch of width shall be not less than that specified in table II.

3.6 Identification marking.—Each sheet of class 1 flint finishing paper and class 2, flint paper shall be marked in a legible manner with the manufacturer's name or trademark of such character that the source of manufacture may be readily determined, and with the name of the product and the grade designation. At least every 3 feet of class 3, roll flint paper shall be marked in a legible manner with the name of the abrasive grain, the grade designation, and the letter E to designate backing weight.

TABLE II.—*Tensile strength (minimum)*

Class No.	Grade	Pounds per inch of width	
		Stronger direction	Weaker direction
1	All grades, 5/0 thru 2/0	20	10
2	Extra fine, fine	35	15
2	Medium, coarse, extra coarse	40	20
3	All grades, extra fine thru extra coarse	85	45

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—At least 3 samples shall be taken from each lot of 1,000 sheets or 10 rolls, or fraction thereof, of each class and grade delivered, for inspection and tests.

4.2 Inspection.—The samples selected shall be inspected with respect to material, workmanship, marking, and dimensional requirements.

4.3 Tests.

4.3.1 *Size of grain.*—The size of the abrasive grain as indicated by the grade designation on the back of each sample sheet or roll shall be in accordance with table III when compared with a series of grits known as "standard sands"[1] for the grading of aluminum oxide and silicon carbide coated abrasive grain. Table III shows the approximate relation between the flint grades and the corresponding grade of artificial abrasive.

TABLE III.—*Comparative grit (grade) numbers*

Silicon carbide and aluminum oxide grit numbers	Class 1 flint grades	Classes 2 and 3 flint grades	Equivalent grade numbers
240	5/0		
220	4/0	Extra fine (XF)	4/0
180	3/0		3/0
150	2/0	Fine (F)	2/0
100 and 120			0
80 and 100		Medium (M)	½
60 and 80			1
50		Coarse (C)	1½
40			2
36		Extra coarse (XC)	2½
30			3

[1] Standard sands for silicon carbide and aluminum oxide grain are obtainable, so far as known, only from the Carborundum Co., Niagara Falls, N. Y.

4.3.2 *Tensile strength.*—At least five strength test pieces 1 inch in width shall be cut both longitudinally and transversely from samples taken at random from the material under inspection. These test pieces shall be conditioned for 24 hours at a temperature of 70° plus or minus 2°F. and 50 plus or minus 2 percent relative humidity. The test pieces, immediately at the end of the conditioning period, shall be placed in a tensile testing machine with the jaws 5 inches apart, and pulled to rupture at the rate of 12 inches per minute. (The required strength is not specified for the longitudinal and transverse directions, respectively, as the sheets are not always cut in the same direction of the paper.)

4.3.3 Methods of test for paper shall be in accordance with Federal Specification UU–P–31.

4.3.4 Material represented by samples shall be accepted or rejected in its entirety based on the results of tests of the representative samples.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Sheet form (classes 1 and 2).*—Flint finishing paper sheets (class 1) and flint

TABLE IV.

Class No.	Grades	Sheets per sleeve or folder	Sleeves or folders per carton	Number of sheets per carton
1	All grades, 5/0 thru 2/0	100	10	1000
2	Extra fine, fine, medium	100	10	1000
2	Coarse, extra coarse	50	10	500

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paper sheets (class 2) may be packed in either chipboard sleeves or folders. The sleeves or folders shall be packed in fiberboard cartons in accordance with table IV.

5.1.2 *Roll form (class 3).*—Unless otherwise specified, roll flint paper (class 3) shall be carefully wrapped in strong paper, and packed in standard commercial containers or wrapping, and shall be prepared for shipment to permit acceptance by carrier for transportation at lowest applicable rate and afford maximum protection from normal hazards of transportation.

5.1.3 *Fiberboard cartons.*—The fiberboard cartons used to package class 1 and class 2, and some forms of class 3 flint paper, shall be in compliance with Federal Specification LLL–B–631 or LLL–B–636. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings, tape, or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification.

5.2 **Marking.** — Individual packages and shipping containers shall be marked with the class, name of the material, the grade designation, and the quantity contained therein, as defined by the contract or order number under which shipment is made, and the name of the manufacturer. In addition, shipping containers shall be marked with the contract or order number.

6. NOTES

6.1 **Ordering data.** — Purchasers should specify the class and grade (or grit number, as applicable) and should exercise any desired options offered herein. (See 1.2.1, 1.2.2, 1.2.3, 3.2, and 5.1.2.)

6.2 **Maximum work value.**—Coated abrasives should be stored in a cool even temperature where the relative humidity is somewhere between 35 and 65 percent. Best work value will be obtained ordinarily if stored and worked in rooms having a relative humidity of 50 percent. It will give a work value of only about 50 percent if used in a 70 percent relative humidity. Coated abrasives should not be stored on a cement floor.

6.2.1 *Belts.*—If coated abrasive belts are worked on wood having excessive moisture, or on glued wood which has not had sufficient time to dry thoroughly, it will cause the belt to fill quickly.

6.3 For comparative grit (grade) numbers, see 4.3.1 (table III).

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards, of the issues in effect on date of invitations for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standards.*

Federal Specifications:

NN–B–591—Boxes; Fiberboard, Wood-Cleated (for Domestic Shipment).

NN–B–601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.

NN–B–621– Boxes; Wood, Nailed and Lock-Corner.

NN–B–631—Boxes; Wood, Wirebound (for Domestic Shipment).

LLL–B–631—Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL–B–636—Boxes; Fiber, Solid (for Domestic Shipment).

Military Specifications:

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

MIL–B–107—Boxes, Wood, Wirebound (Overseas Type).

JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

MIL–P–116—Preservation, Methods of.

JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *For domestic shipment and storage and overseas shipment.*

7.2.2.1.1 *Sheet form (classes 1 and 2).*— Flint finishing-paper sheets (class 1) and flint paper sheets (class 2) shall be packaged in either chipboard sleeves or folders. Number of sheets per sleeve or per folder shall be as follows:

Class No.	Grades	Sheets per sleeve or folder
1	All grades, 5/0 thru 2/0	100
2	Extra fine, fine, medium	100
2	Coarse, extra coarse	50

7.2.2.1.2 *Roll form (class 3).* — Roll flint paper shall be unit packaged in cylinder or stack form by placing rolls successively one upon another. Maximum height of the cylinder or stack shall be 7½ inches. Rolls of 4 or more inches in width shall be individually packaged.

7.2.2.2 Packaging shall be in accordance with the requirements of method IA–2 or IA–8 of Military Specification MIL–P–116. Package quantities shall be as required by 7.2.2.1.

7.2.2.3 *Intermediate packaging.* — Unit packages of rolls and sheets shall be intermediate packaged when 20 or more unit pack-

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ages can be included without exceeding a gross weight of 20 pounds, with a maximum permissible overweight of 3 pounds. Intermediate containers shall conform to Military Specification JAN–P–108 and appendix thereto; or to Federal Specification LLL–B–631 or LLL–B–636, sealed with adhesive, and then packaged according to method IA–2 of Military Specification MIL–P–116.

7.2.3 *Packing.*

7.2.3.1 *For domestic shipment and storage.*—Paper, packaged as described in 7.2.2.1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, except that when wood-cleated fiberboard boxes, wood-cleated plywood boxes, nailed wood boxes, wirebound wood boxes, corrugated fiber boxes, or solid fiber boxes are used, they shall conform to the requirements of Federal Specifications NN–B–591, NN–B–601, NN–B–621, NN–B–631, LLL–B–631, and LLL–B–636, respectively. The weight of wood boxes and contents shall not exceed approximately 200 pounds; and of fiberboard boxes and contents, approximately 90 pounds.

7.2.3.2 *For overseas shipment.* — Paper, packaged as described in 7.2.2.2, shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes, conforming to the requirements of Military Specifications JAN–P–105, JAN–P–106, MIL–B–107, JAN–P–108 compliance symbol V2s, and JAN–P–138, respectively. The weight of wood boxes and contents shall not exceed approximately 150 pounds; and of fiberboard boxes and contents, approximately 70 pounds.

7.2.4 *Marking.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.5 *Ordering data.* — Procurement documents should specify whether domestic or overseas packing and marking are required (see 7.2.3 and 7.2.4).

7.3 Navy.

7.3.1 *Applicable Navy Department Specification:*

General Specifications for Inspection of Material.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3.2 *Inspection procedures.* — For Navy purchases the general inspection procedures shall be in accordance with the General Specifications for Inspection of Material.

7.3.3 *Sampling for lot acceptance and tests.*

7.3.3.1 *Inspection lot.* — All paper of the same grit number presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

7.3.3.2 *Sampling for inspection.*

7.3.3.2.1 A random sample of paper shall be taken from each inspection lot of material offered for Government inspection of visual and dimensional characteristics, with lot acceptance based on the sampling inspection requirements in accordance with Military Standard MIL–STD–105. The Acceptable Quality Level shall be 2.5 percent defective.

7.3.3.2.2 Unless otherwise specified by the procuring agency, Inspection Level I and multiple sampling shall be utilized.

7.3.4 *Visual and dimensional inspection and tests.* — Each of the sample sheets selected in accordance with 7.3.2.2 shall be visually and dimensionally inspected and tested by the Government inspector to verify compliance with the requirements of this

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specification. Any sheet in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective sheets in any sample exceeds the acceptance number for the sample specified by Military Standard MIL–STD–105, the lot represented by the sample shall be rejected.

NAVY INTEREST: *MC* Or Sh S Y

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